

SONiX 32-Bit Cortex-M0 Micro-Controller

RTC Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of RTC.

2 FEATURE

- Second Mode : Generate an interrupt every second.
- Alarm Mode : Generate the alarm interrupt at 10 seconds after starting the RTC.
- Overflow Mode : Generate the overflow interrupt at 15.271 hours after starting the RTC.
- Use ARM-compiler V6.12

★ ***Note: Only SN32F700/100/760/240 support Alarm mode and Overflow mode.***

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Second Mode

1. Open RTC.h, and set RTC_MOD as SECOND.

```
1 #ifndef __SN32F760_RTC_H
2 #define __SN32F760_RTC_H
3
4
5 /*_____ I N C L U D E S _____*/
6 #include <SN32F760.h>
7
8 /*_____ D E F I N I T I O N S _____*/
9 #define SECOND      0
10 #define ALARM       1
11 #define OVERFLOW    2
12 #define RTC MODE    SECOND      //SECOND, ALARM, OVERFLOW
```

2. Call RTC_Init();

```
int main (void)
{
    //init RTC
    RTC_Init();

    .
    .
    User code
    .
    .
}
```

3. Modify the settings inside the following red circle to set the RTC clock source and Second counter reload value.

Example: Set RTC clock source as RTC_CLKSEL_ELS,
 Second counter reload value = 32767

```

void RTC_Init(void)
{
    __RTC_ENABLE_RTCHCLK;           //Enable HCLK for RTC

    #if RTC_MODE == SECOND
        RTC_SelectClockSource(RTC_CLKSEL_ELS); //Second will occur every 1 second
        SN_RTC->IE = mskRTC_SECIE_ENABLE; //Clock Source select
        __RTC_SECCNTV(32767); //Enable Second Interrupt
        //Second counter reload value
    #endif

    #if RTC_MODE == ALARM
        RTC_SelectClockSource(RTC_CLKSEL_ELS); //Alarm will occur after 10 seconds
        SN_RTC->IE = mskRTC_ALMIE_ENABLE; //Clock Source select
        __RTC_SECCNTV(32767); //Enable Alarm Interrupt
        __RTC_ALMCNTV(9); //Second counter reload value
        //Alarm counter reload value
    #endif

    #if RTC_MODE == OVERFLOW
        RTC_SelectClockSource(RTC_CLKSEL_EHS); //Overflow will occur in 54975.58139 seconds(15.271 hours)
        SN_RTC->IE = (mskRTC_OVFIE_ENABLE | mskRTC_ALMIE_ENABLE); //Clock Source select
        __RTC_SECCNTV(1); //Enable Overflow Interrupt
        __RTC_ALMCNTV(75000000); //Second counter reload value & Second will occur in 12.8u second
        //Alarm counter reload value & Alarm will occur in 960 seconds(16 minutes)
    #endif

    //Enable RTC NVIC interrupt
    RTC_NvicEnable();

    __RTC_ENABLE; //Enable RTC
}

```

4.2 Alarm Mode

★ **Note: Only SN32F700/100/760/240 support Alarm mode and Overflow mode.**

1. Open RTC.h, and set RTC_MOD as ALARM.

```

1 #ifndef __SN32F760_RTC_H
2 #define __SN32F760_RTC_H
3
4
5 /*_____ I N C L U D E S _____*/
6 #include <SN32F760.h>
7
8 /*_____ D E F I N I T I O N S _____*/
9 #define SECOND      0
10 #define ALARM       1
11 #define OVERFLOW    2
12 #define RTC_MODE    ALARM //SECOND, ALARM, OVERFLOW

```

2. Call RTC_Init();

```

int main(void)
{
    //init RTC
    RTC_Init();

    .
    .
    User code
    .
    .
}

```

3. Modify the settings inside the following red circle to set the RTC clock source, Second counter reload value and Alarm counter reload value.

Example: Set RTC clock source as RTC_CLKSEL_ELS
 Second counter reload value = 32767
 Alarm counter reload value = 9

```
void RTC_Init(void)
{
    __RTC_ENABLE_RICHCLK;           //Enable HCLK for RTC

    #if RTC_MODE == SECOND
        RTC_SelectClockSource(RTC_CLKSEL_ELS);           //Second will occur every 1 second
        SN_RTC->IE = mskRTC_SECIE_ENABLE;                //Clock Source select
        __RTC_SECCNTV(32767);                             //Enable Second Interrupt
        __RTC_ALMCNTV(9);                                //Second counter reload value
    #endif

    #if RTC_MODE == ALARM
        RTC_SelectClockSource(RTC_CLKSEL_ELS);           //Alarm will occur after 10 seconds
        SN_RTC->IE = mskRTC_ALMIE_ENABLE;                //Clock Source select
        __RTC_SECCNTV(32767);                             //Enable Alarm Interrupt
        __RTC_ALMCNTV(9);                                //Second counter reload value
    #endif

    #if RTC_MODE == OVERFLOW
        RTC_SelectClockSource(RTC_CLKSEL_EHS);           //Alarm counter reload value
        SN_RTC->IE = (mskRTC_OVFIE_ENABLE | mskRTC_ALMIE_ENABLE); //Overflow will occur in 54975.58139 seconds(15.271 hours)
        __RTC_SECCNTV(1);                                //Clock Source select
        __RTC_ALMCNTV(75000000);                         //Enable Overflow Interrupt
    #endif
    //Second counter reload value & Second will occur in 12.8u second
    //Alarm counter reload value & Alarm will occur in 960 seconds(16 minutes)

    //Enable RTC NVIC interrupt
    RTC_NvicEnable();

    __RTC_ENABLE;           //Enable RTC
}
```

4.3 Overflow Mode

★ **Note: Only SN32F700/100/760/240 support Alarm mode and Overflow mode.**

1. Open RTC.h, and set RTC_MOD as OVERFLOW.

```
1 #ifndef __SN32F760_RTC_H
2 #define __SN32F760_RTC_H
3
4
5 /*_____ I N C L U D E S _____*/
6 #include <SN32F760.h>
7
8 /*_____ D E F I N I T I O N S _____*/
9 #define SECOND      0
10 #define ALARM       1
11 #define OVERFLOW    2
12 #define RTC_MODE    OVERFLOW           //SECOND, ALARM, OVERFLOW
```

```

2. Call RTC_Init();
int main (void)
{
    //init RTC
    RTC_Init();

    .
    .
    User code
    .
    .
}

```

3. Modify the settings inside the following red circle to set the RTC clock source, Second counter reload value and Alarm counter reload value.

Example : Set RTC clock source as RTC_CLKSEL_EHS

- EHS X'TAL = 20MHz
- When the Clock source is set to EHS, please refer to the instructions in section 3.1

Second counter reload value = 1

Alarm counter reload value = 75000000

```

void RTC_Init(void)
{
    __RTC_ENABLE_RTCHCLK;                //Enable HCLK for RTC

    #if RTC_MODE == SECOND                //Second will occur every 1 second
    RTC_SelectClockSource(RTC_CLKSEL_ELS); //Clock Source select
    SN_RTC->IE = mskRTC_SECIE_ENABLE;     //Enable Second Interrupt
    __RTC_SECCNTV(32767);                 //Second counter reload value

    #endif

    #if RTC_MODE == ALARM                 //Alarm will occur after 10 seconds
    RTC_SelectClockSource(RTC_CLKSEL_ELS); //Clock Source select
    SN_RTC->IE = mskRTC_ALMIE_ENABLE;     //Enable Alarm Interrupt
    __RTC_SECCNTV(32767);                 //Second counter reload value
    __RTC_ALMCNTV(9);                     //Alarm counter reload value

    #endif

    #if RTC_MODE == OVERFLOW              //Overflow will occur in 54975.58139 seconds(15.271 hours)
    RTC_SelectClockSource(RTC_CLKSEL_EHS); //Clock Source select
    SN_RTC->IE = (mskRTC_OVFIE_ENABLE | mskRTC_ALMIE_ENABLE); //Enable Overflow Interrupt
    __RTC_SECCNTV(1);                     //Second counter reload value & Second will occur in 12.8u second
    __RTC_ALMCNTV(75000000);               //Alarm counter reload value & Alarm will occur in 960 seconds(16 minutes)
    #endif

    //Enable RTC NVIC interrupt
    RTC_NvicEnable();

    __RTC_ENABLE;                         //Enable RTC
}

```


3.1 Note

- Modify the settings inside the following red circle depend on EHS X'TAL.
- When the EHS X'TAL $\leq$ 12MHz, set to SYS0_EHS_FREQ_DRIVE_LOW
- When the EHS X'TAL $>$ 12MHz, set to SYS0_EHS_FREQ_DRIVE_HIGH

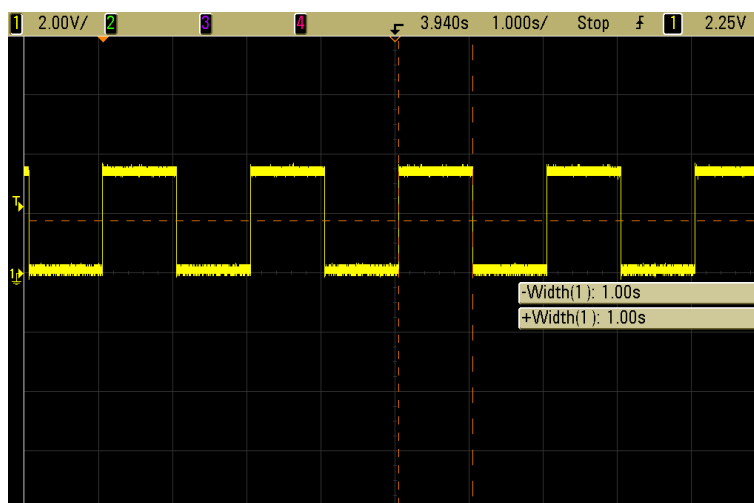
```
void RTC_SelectClockSource(uint32_t src)
{
    if (src == RTC_CLKSEL_ELS)
        SYS0_EnableELSXtal();
    else if (src == RTC_CLKSEL_EHS)
        SYS0_EnableEHSXtal(SYS0_EHS_FREQ_DRIVE_HIGH);

    SN_RTC->CLKS = src;                                     //clock source select
}
```

5 RESULT

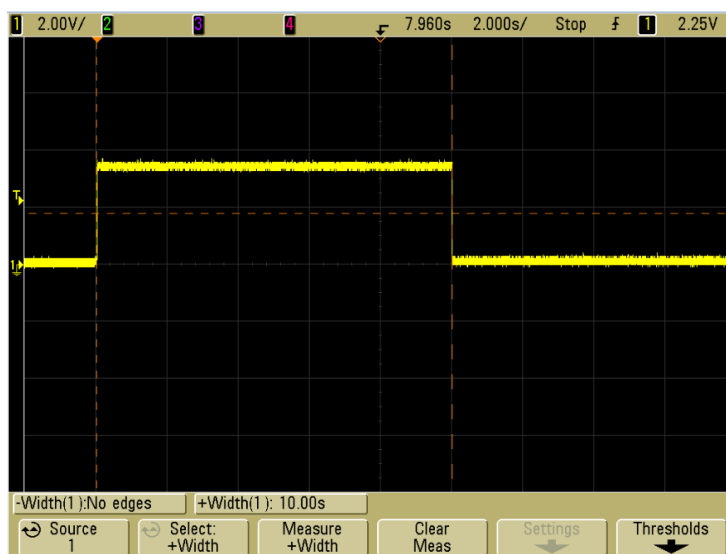
5.1 Second Mode

The RTC is configured to generate an interrupt every second, and P1.0 toggles every second.



5.2 Alarm Mode

The RTC is configured to generate an alarm interrupt 10 seconds after RTC starts counting. P1.1 is high when RTC starts to count, and goes low when the RTC alarm interrupt is generated.



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